Smart Load-Sharing EV Charging Station with Thermal Protection

Intelligent Multi-EV Charging Controller

Power Electronics Course Project (MATLAB/Simulink simulation)
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Abstract

This report presents the design and simulation of a smart load-sharing electric-vehicle (EV) charging station in MATLAB/Simulink. Two EV batteries charge at the same time from a single source limited to 50 A. Each battery is charged through its own buck converter using constant-current / constant-voltage (CC-CV) control. A load-sharing controller scales the two current requests so the total never exceeds the limit, and a Stateflow thermal-protection state machine derates or pauses a charger when its simplified battery temperature model crosses 45 °C or 55 °C. If one EV pauses, the freed current is automatically available to the other.

Against a fixed-current baseline charger with no load sharing and no thermal protection, the smart controller holds the total charging current at the 50 A limit (the baseline draws about 70 A) and keeps the EV1 battery near 45.5 °C (the baseline reaches about 87 °C). In an overheating scenario, EV1 pauses and resumes repeatedly while EV2 keeps charging.

1. Introduction

A home or office charging point usually has a limited supply, set by the breaker or the grid connection. When two vehicles are plugged in at once, simply giving each one its full charging current can overload that supply. Charging also heats the battery, and a charger that ignores temperature can push a battery beyond its safe range. This project addresses both problems with a controller that (1) splits the available power between two EVs, (2) never exceeds the supply limit, and (3) slows down or pauses charging when a battery overheats.

The objectives of the project were to:

1. Design a Simulink model of a shared charging station where two EVs charge simultaneously from a single power-limited source.
2. Implement CC-CV charging control for each EV battery.
3. Develop a power-sharing algorithm that dynamically allocates the available current between the two EVs without exceeding the grid limit.

4. Add a simplified thermal model for each battery and a protection logic that reduces or pauses charging above a safe temperature.
5. Compare the system against a baseline charger with no load sharing or thermal protection.

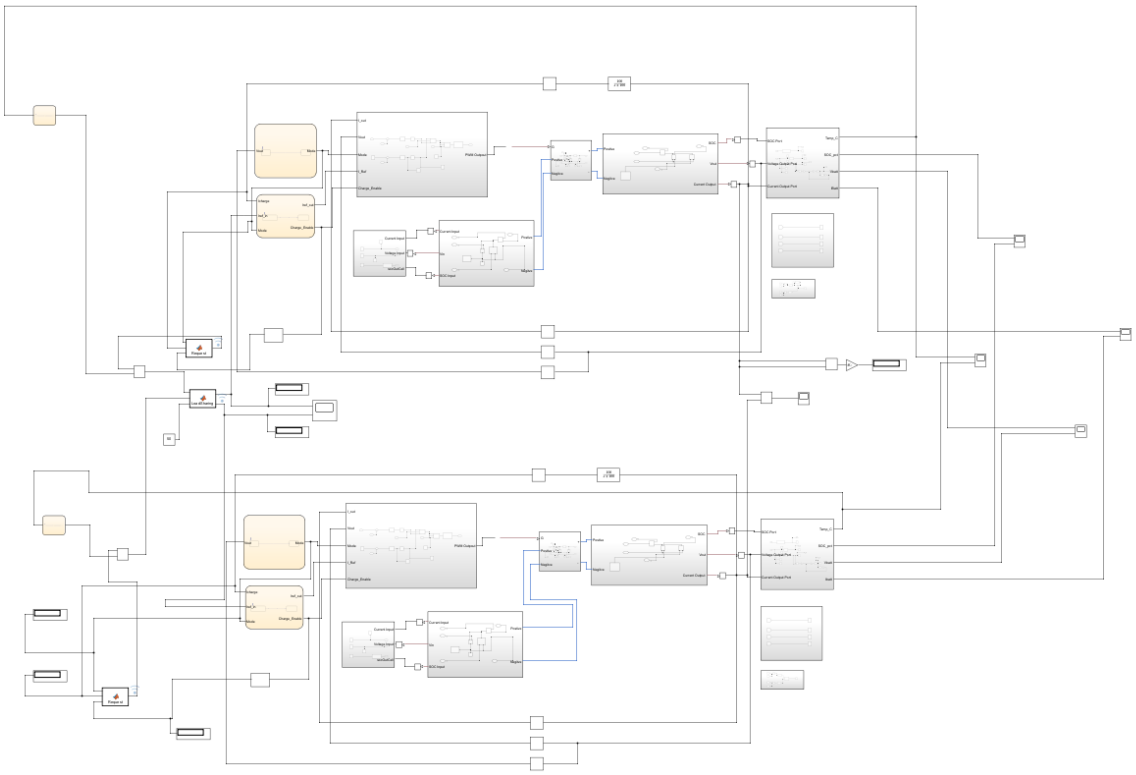
CC-CV charging on its own is a standard topic. The contribution of this project is the combination of multi-EV load sharing with thermal protection, and in particular their interaction: when one EV is derated or paused for heat, the load sharing passes the spare current to the other EV.

2. System Description

2.1 Architecture

The station consists of two identical charging channels, EV1 and EV2. Each channel contains an input source, a buck converter, a battery, a CC-CV controller with mode logic, a current-termination block, and measurement blocks. Three shared elements sit above the two channels: the load-sharing controller, a request block for each EV, and a thermal-protection state machine for each EV.

The signal flow for each EV is as follows. The request block states how much current the EV wants. The thermal-protection block multiplies that request by a derating factor. The load-sharing controller then limits the two requests so that their sum stays within the supply limit, and returns a current reference to each channel. Each channel regulates its battery current to that reference.



2.2 Model parameters

Table 1 lists the main parameters of the model. Items highlighted in yellow were not available when this draft was written and should be filled in from the Simulink model.

Block	Parameter	Value
Battery (each EV)	Nominal voltage	350 V
	Capacity	309 A·hr
	Internal resistance	0.048 Ω
	Voltage at 61.8 A·hr (no load)	336 V
	Initial charge	15.45 A·hr (about 5 % SOC)
Buck converter	Switch	Ideal switching IGBT: Vth 0.5 V, Vf 0.8 V, Ron 1 mΩ
	Diodes	Vf 0.8 V, Ron 1 mΩ, off conductance 1e-5 S
	Input voltage	430 V nominal
	Inductor	0.0037 L
	Output capacitor	1e-5 C
	Switching frequency	1e-4
CC-CV controller	Current loop (discrete PI)	Kp = 0.2, Ki = 10
	Voltage loop (discrete PI)	Kp = 15, Ki = 0.1
	Voltage reference	295 V
	CC to CV transition	Vout ≥ 294.35 V
Measurements	Low-pass filter on V and I	100 / (s + 100)
Load sharing	Supply current limit Imax	50 A
	Request per EV in CC stage, Icc	35 A
Thermal model	Heating resistance	0.05 Ω
	Heat capacity C_th	0.25 J/K
	Thermal resistance to ambient R_th	1.33 K/W
	Ambient temperature	25 °C
Thermal protection	Slow-down / pause thresholds	45 °C / 55 °C
	Resume thresholds	50 °C (to slow-down), 40 °C (to full)
Simulation	Solver / stop time	ode23t (variable step) / 0.5 s

Table 1: Main model parameters

3. Control Strategy

3.1 CC-CV charging

Each charger uses two nested discrete PI loops. A voltage loop compares the battery voltage with the 295 V reference and produces a current demand, and a current loop compares the battery current with the current reference and produces the duty cycle of the buck converter. A Stateflow chart, CC-CV

Logic, sets a Mode signal: it is 0 (constant current) at the start and changes to 1 (constant voltage) once the battery voltage reaches 294.35 V. A switch in the controller selects the current loop in CC mode and the voltage loop in CV mode. A current-termination block disables charging through a Charge_Enable signal at the end of charge.

3.2 Load sharing

The load-sharing controller is a MATLAB Function block. It takes the requested currents Ireq1 and Ireq2 and the supply limit Imax. If the total request is within the limit, both requests pass through unchanged. If it is over the limit, both are scaled down by the same factor:

$$k = Imax / (Ireq1 + Ireq2), \quad Iref1 = k \cdot Ireq1, \quad Iref2 = k \cdot Ireq2 \quad (when\ Ireq1 + Ireq2 > Imax)$$

Proportional scaling gives each EV a share in proportion to what it asked for, so equal requests give an equal split. It also means that if one EV asks for less, or for nothing, the other EV receives up to its full request, limited only by Imax. This is what lets spare current pass automatically from one EV to the other. The code is given in Appendix A.

3.3 Current request

A request block for each EV decides what to ask for, based on that EV's state. If charging is disabled (Charge_Enable is 0), the request is 0 A. In the constant-current stage, the request is the full 35 A. In the constant-voltage stage, the request equals the current the battery is actually drawing, which falls as the battery fills. Using the actual current in the CV stage matters: a fixed request would keep reserving 35 A of the supply for a battery that is only drawing a few amps, and the other EV could not use the difference.

The Charge_Enable signal passes through a one-sample delay before it reaches the request block. This breaks an algebraic loop (Charge_Enable → request → load sharing → current reference → termination logic → Charge_Enable) that Simulink cannot solve.

3.4 Battery thermal model

Each battery has a first-order lumped thermal model built from basic blocks. Heat is generated by the battery current through an effective resistance, and heat leaves through a thermal resistance to ambient:

$$C_th \cdot dT/dt = I^2 \cdot R_heat - (T - T_amb) / R_th$$

With R_heat = 0.05 Ω, C_th = 0.25 J/K, R_th = 1.33 K/W and T_amb = 25 °C, a charging current of 25 A gives about 31 W of heating and a steady-state temperature near 67 °C if nothing intervenes. The equation is implemented with a Product block (I^2), a Gain (0.05), a Sum block, a Gain (1/C_th = 4), an Integrator (initial value 25 °C), and a feedback path through a Bias (−25) and a Gain (0.75).

The parameters are scaled so that the temperature changes noticeably within the 0.5 s simulation. A real 309 A·hr battery has a far larger heat capacity and takes hours to warm up. The Simscape battery block also contains its own thermal port with a 30,000 J/K thermal mass; with that value the temperature rose by only about 0.0005 °C during a 0.5 s run, so it was not suitable for demonstrating the protection logic.

3.5 Thermal protection

A Stateflow chart for each EV reads the battery temperature and outputs a derating factor, Derate. The request is multiplied by Derate before it enters the load-sharing controller. The chart has three states:

State	Derate	Enters when
Charging	1.0 (full)	Start; or temperature < 40 °C while slowed down
SlowingDown	0.5 (half)	Temperature > 45 °C while charging; or temperature < 50 °C while paused
Paused	0 (stopped)	Temperature > 55 °C while slowed down

Table 2: Thermal protection states

The resume thresholds are lower than the trip thresholds. This hysteresis stops the chart from switching back and forth when the temperature sits close to a limit. Paused returns to SlowingDown first, and not directly to Charging, so the current comes back in two steps.

4. Baseline Charger

The baseline uses the same two charging channels, batteries and thermal calculation, but removes everything added for the smart strategy: the load-sharing controller, the request blocks, the Derate multiplication and both thermal-protection charts. Each EV receives a fixed 35 A current setpoint. The thermal model is kept in the baseline only to measure battery temperature, so the two systems can be compared on the same scale.

5. Simulation Scenarios

All results come from 0.5 s simulations. The following cases were run.

- Scenario 1 and 3, normal charging with a grid-limit split: both EVs have identical settings and each requests 35 A. The combined 70 A is above the 50 A limit, so the load sharing has to split the current.
- Scenario 2, one EV overheating: the effective heating resistance of EV1 is tripled, from 0.05 Ω to 0.15 Ω, to emulate a hotter battery. EV2 is unchanged.
- Baseline: both EVs at a fixed 35 A with no protection, for comparison with the normal case.
- Functional checks: an EV starting at 95 % SOC (to force it into CV mode), and Charge_Enable forced to 0 for EV1 (to emulate a pause).

6. Results

6.1 Normal charging and grid-limit split

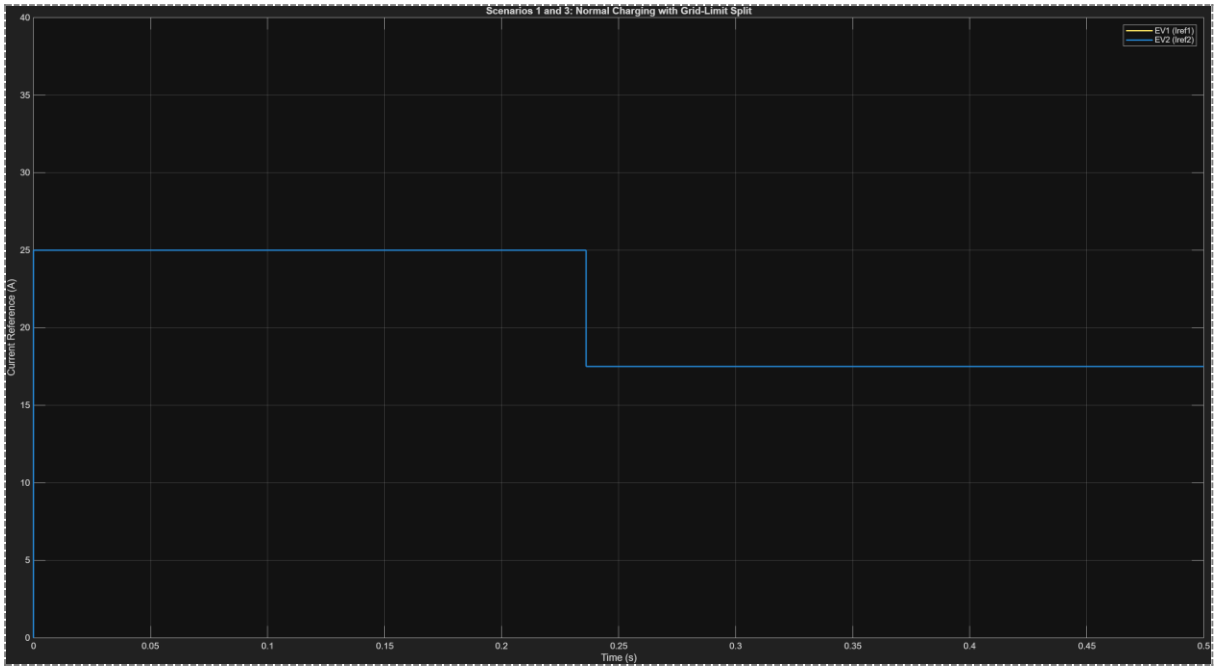


Figure 1: Current references for EV1 and EV2 in normal charging. The two lines overlap exactly.

Both EVs request 35 A, so the controller scales each reference to 25 A, which keeps the total at 50 A. At about 0.236 s both batteries reach 45 °C, enter the SlowingDown state, and request 17.5 A each. The total request of 35 A is now below the limit, so no scaling is needed. The two lines coincide because the EVs are identical.

6.2 Total grid current: smart controller and baseline

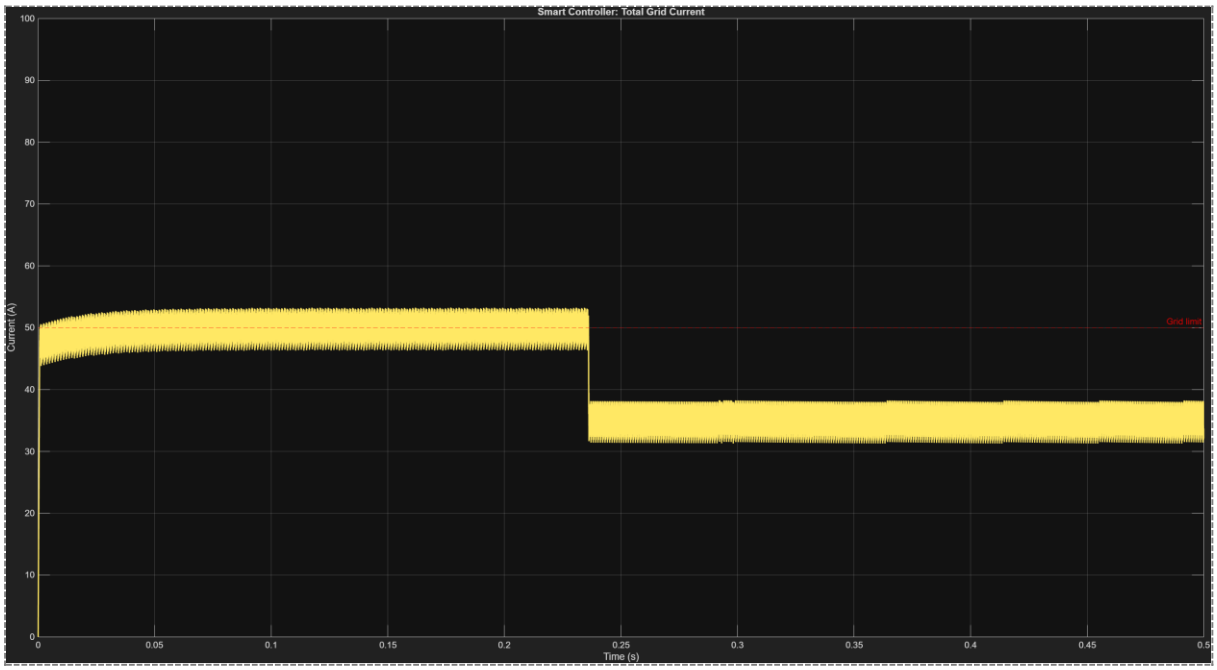


Figure 2: Total charging current with the smart controller, against the 50 A limit.

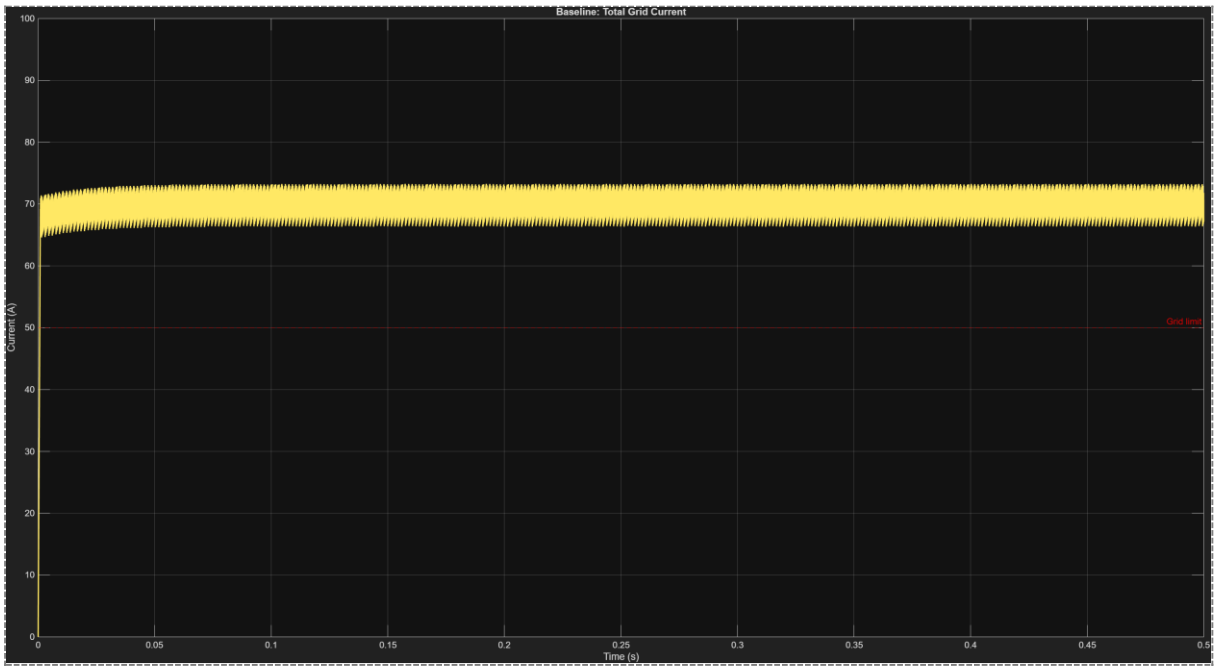


Figure 3: Total charging current with the baseline charger, against the 50 A limit.

With the smart controller the average total current sits at the 50 A limit until about 0.236 s, then falls to about 35 A when both EVs slow down for heat. The switching ripple on the measured current reaches peaks of about 53 A. The baseline draws about 70 A for the whole run, 20 A (40 %) above the limit.

6.3 Battery temperature: smart controller and baseline

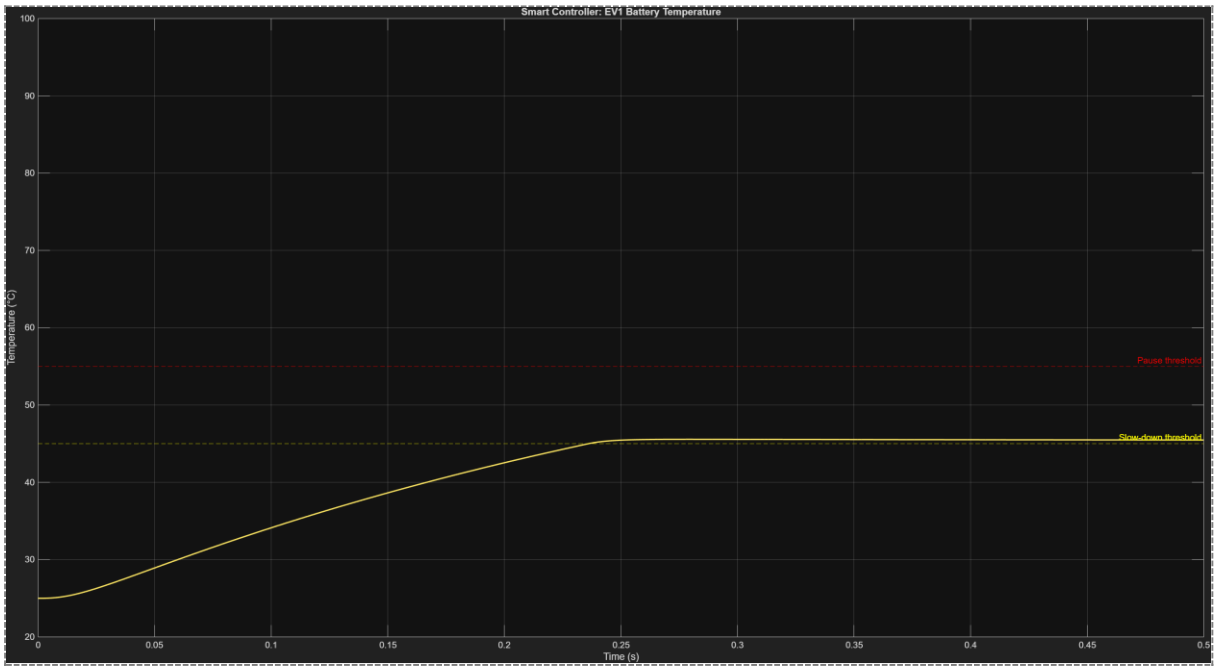


Figure 4: EV1 battery temperature with the smart controller.

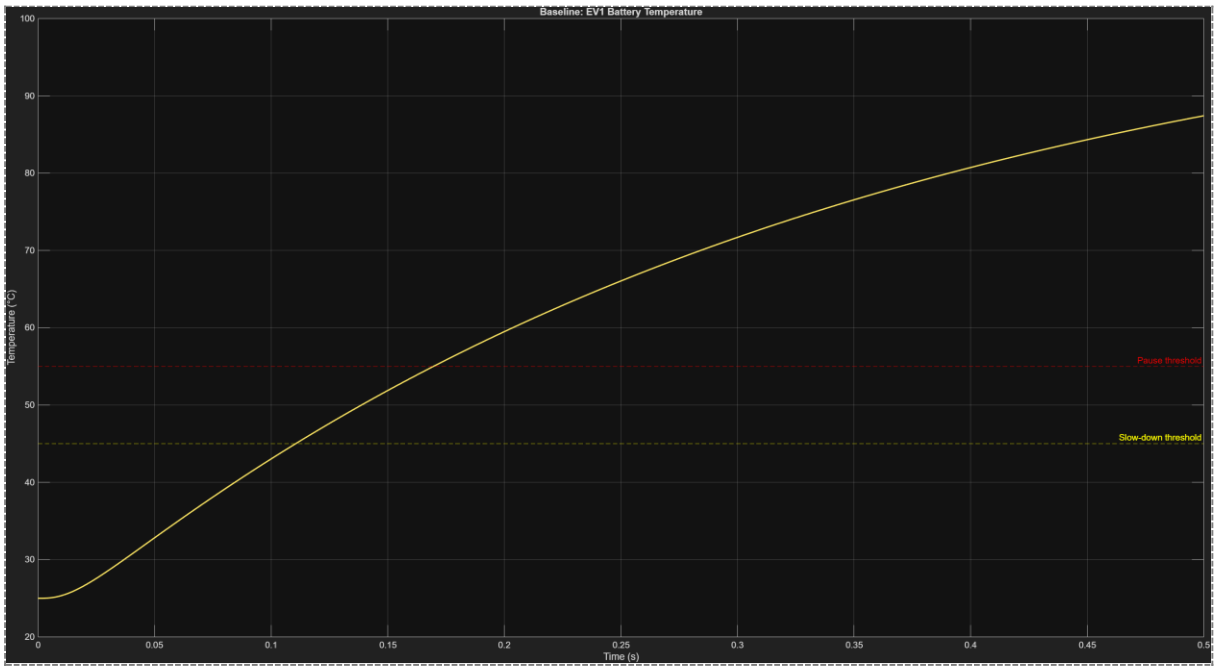


Figure 5: EV1 battery temperature with the baseline charger.

With the smart controller the battery temperature rises to 45 °C at about 0.236 s. The derated current then holds it at about 45.5 °C, well under the 55 °C pause threshold. In the baseline, the temperature crosses 45 °C at about 0.11 s and 55 °C at about 0.17 s and keeps rising, reaching about 87 °C at 0.5 s.

6.4 One EV overheating

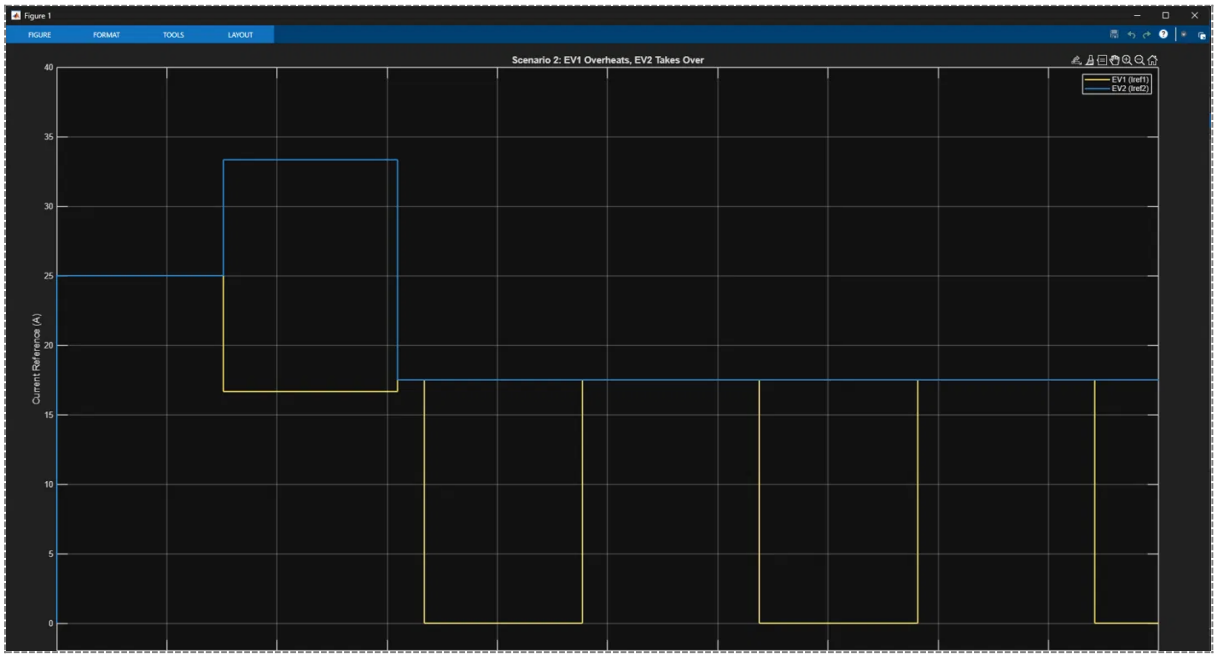


Figure 6: Current references when EV1 overheats. EV1 slows down and then pauses repeatedly.

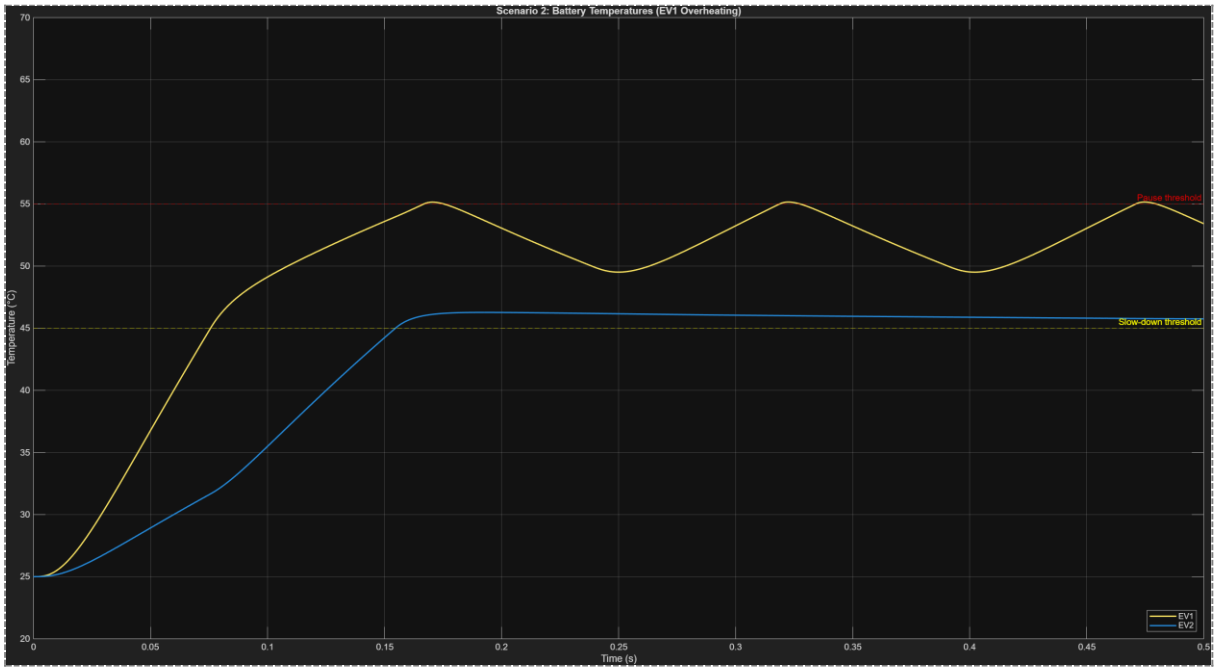


Figure 7: Battery temperatures in the overheating scenario.

EV1 heats faster than EV2. The sequence of events was:

- At about 0.075 s, EV1 reaches 45 °C and slows down. Its reference falls to about 16.7 A and EV2's rises to about 33 A.
- At about 0.155 s, the extra current has heated EV2 past 45 °C, so EV2 slows down too and falls to 17.5 A.
- At about 0.165 s, EV1 reaches 55 °C and pauses. Its reference drops to 0 A while EV2 continues at 17.5 A.
- At about 0.24 s, EV1 has cooled to about 50 °C and resumes at the reduced rate. It heats back to 55 °C and pauses again at about 0.32 s and 0.47 s.

EV2's temperature settles near 46 °C. This shows that EV1 is repeatedly stopped by its own temperature while the other EV keeps charging. It also shows that the two protections work independently: the current EV1 gives up is available to EV2, but EV2 is still limited by its own temperature.

6.5 Battery voltage, current and SOC

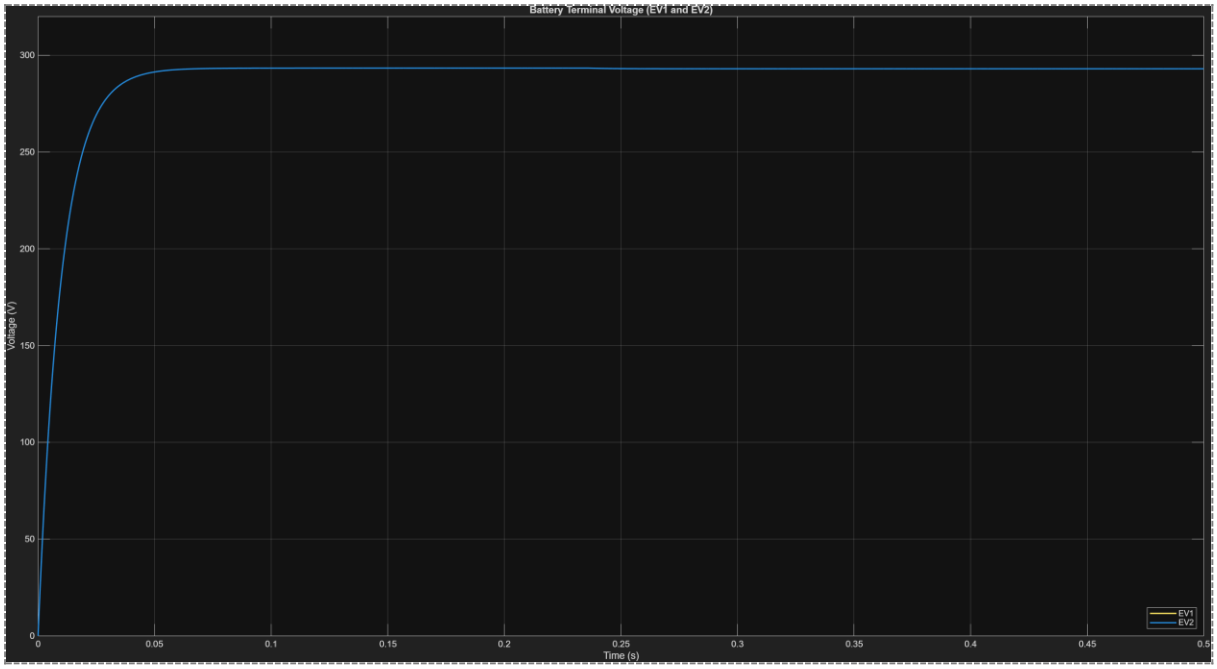


Figure 8: Battery voltage of EV1 and EV2.

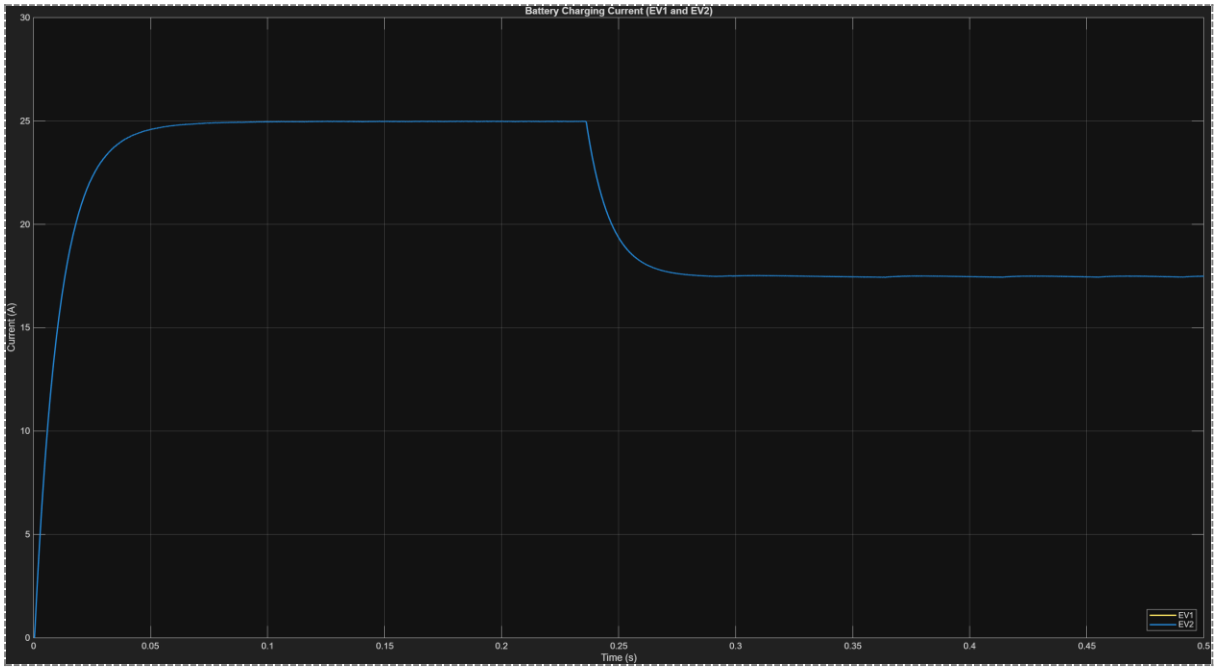


Figure 9: Battery charging current of EV1 and EV2 (filtered).

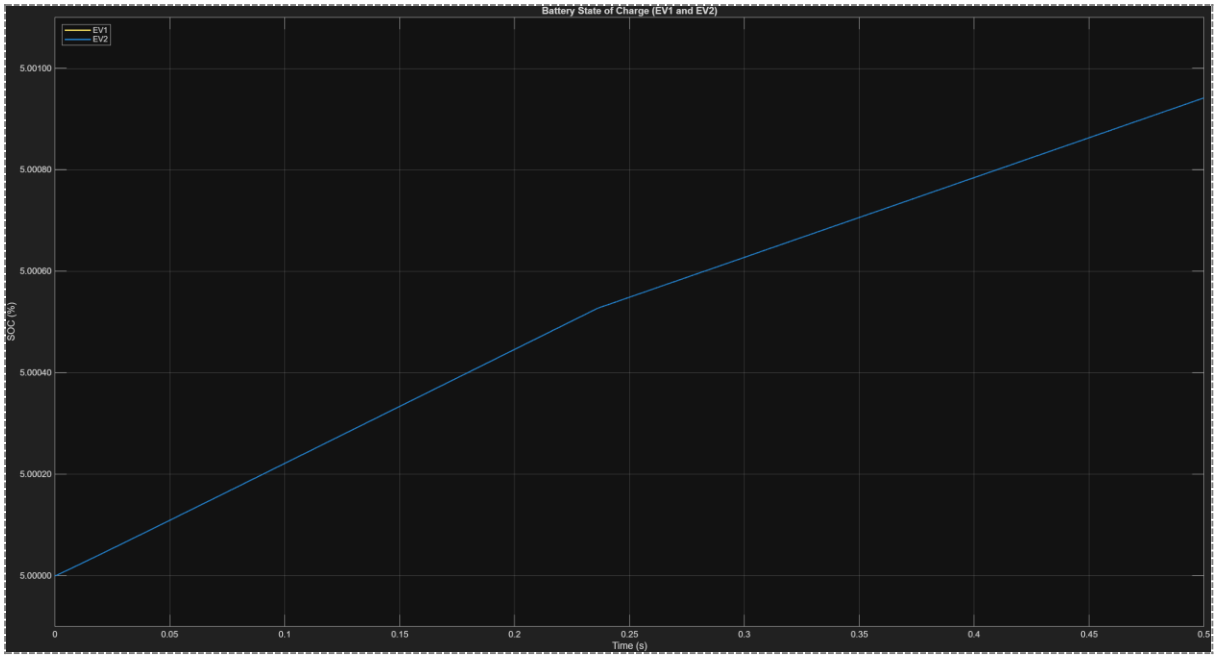


Figure 10: State of charge of EV1 and EV2.

The battery voltage rises over about 0.06 s to roughly 293 V, just under the 294.35 V CV threshold, so the batteries remain in the CC stage during the run. The battery current rises to 25 A in the same time and then falls to 17.5 A after 0.236 s, following the current references. SOC rises from 5 % to about 5.0009 %, with the slope reducing after 0.236 s when the current is derated. The change is small because 0.5 s is very short compared with the hours needed to charge a 309 A·hr battery; the slope change is the useful feature.

6.6 Functional checks

- With EV1 starting at 95 % SOC, EV1 goes into CV mode almost immediately and its request falls to about 0 A. EV2 then receives 35 A instead of 25 A.
- With EV1's Charge_Enable forced to 0, its reference is 0 A and EV2 receives 35 A.

Both checks confirm that spare current passes from one EV to the other without any extra logic.

6.7 Summary

Metric	Baseline	Smart controller
Total charging current	About 70 A throughout	About 50 A, then about 35 A after 0.236 s
Supply limit (50 A) respected	No, exceeded by about 20 A	Yes on average; ripple peaks about 53 A
EV1 temperature at 0.5 s	About 87 °C	About 45.5 °C
Time to reach 55 °C	About 0.17 s	Not reached
Response to overheating	None	Slow down at 45 °C, pause at 55 °C, resume at 50 °C
Spare current from a paused EV	Not shared	Passed to the other EV automatically

Table 3: Baseline and smart controller compared

7. Discussion and Limitations

- Time scaling: the thermal parameters are deliberately scaled so heating happens within 0.5 s. The temperature values and times show how the controller behaves, but they are not a prediction of real charging times or real battery temperatures.
- Supply limit measurement: the limit is applied to the battery-side current, not to the current drawn from the grid, and the converter losses are not included. Switching ripple makes the instantaneous total exceed 50 A by about 3 A even though the average is at the limit. A margin on Imax, or a more heavily filtered measurement, would remove this.
- Charging time: derating the current makes the smart controller slower than the baseline when a battery is hot, because it runs at 17.5 A instead of 25 A or more. This is the trade-off for protecting the battery. The 0.5 s run is too short to measure complete charging times, so only the current and temperature behaviour are compared.
- Identical EVs: in the normal scenarios both EVs have the same settings, so their curves overlap. Only the overheating scenario treats them differently.
- Thermal model: it is a single lumped, first-order model with a constant heating resistance. It does not include ambient changes, cooling systems, or the dependence of battery resistance on temperature or SOC.
- Hysteresis cycling: in scenario 2, EV1 pauses and resumes repeatedly. The 5 °C hysteresis band prevents rapid switching, but a real controller might also hold the pause for a minimum time or latch it until the temperature has fallen further.
- Fairness: proportional scaling is a simple policy. Other policies, such as priority for the EV with the lower SOC, could be added without changing the structure.

8. Conclusion

A Simulink model of a two-EV charging station with CC-CV control, proportional load sharing and Stateflow thermal protection was built and tested. The load-sharing controller kept the combined charging current at the 50 A limit, where the fixed-current baseline drew about 70 A. The thermal protection held the EV1 battery at about 45.5 °C, where the baseline reached about 87 °C. When EV1 was made to overheat, it paused and resumed repeatedly while EV2 continued to charge, and the current EV1 gave up was passed to EV2 automatically. That interaction between the load sharing and the thermal protection is the key result of the project. Possible extensions include grid-side current measurement, priority-based sharing, a more detailed thermal model, and simulation of a full charge.

Appendix A: MATLAB Function Blocks

Load sharing controller:

```
function [Iref1, Iref2] = LoadSharing(Ireq1, Ireq2, Imax)
% Ireq1/2 : current each EV requests (A)
% Imax    : total current allowed by the supply (A)
% Iref1/2  : current each charger is allowed to draw (A)

total = Ireq1 + Ireq2;

if total > Imax && total > 0
   k = Imax / total;      % proportional scale-down
   Iref1 = Ireq1 * k;
   Iref2 = Ireq2 * k;
else
   Iref1 = Ireq1;         % under the limit, no change
   Iref2 = Ireq2;
end
```

```
end
```

Current request:

```
function Ireq = Request(Mode, Icharge, Charge_Enable)
% Mode = 0 -> CC stage: ask for the full 35 A
% Mode = 1 -> CV stage: ask for the current actually drawn
% Charge_Enable = 0 -> charging stopped: ask for 0 A
Icc = 35;

if Charge_Enable < 0.5
   Ireq = 0;
elseif Mode < 0.5
   Ireq = Icc;
else
   Ireq = Icharge;
end
end
```